Enterprise LAN Network Architecture: Secure Multi-VLAN Inter-VLAN Routing & Traffic Control

1. Project Overview, Tools & Protocols

Project Overview

This project demonstrates the design, implementation, and troubleshooting of a secure enterprise Local Area Network (LAN) simulated in **Cisco Packet Tracer**. The architecture focuses on Layer 2 security, scalable Inter-VLAN routing via Router-on-a-Stick, centralized DHCP provisioning, and directional traffic enforcement using Extended Access Control Lists (ACLs).

Simulation & Hardware Tools

- **Cisco Packet Tracer:** The primary simulation platform used for designing, cabling, configuring, and verifying the network topology.
- **Cisco 2911 Router (`R1`):** Acts as the core layer-3 gateway performing Inter-VLAN routing, hosting the central DHCP server, and enforcing security policies via ACLs.
- **Cisco 2960 Switch (`SW1`):** Manages Layer 2 switching, VLAN segmentation, trunking, and port security features.
- **End-Devices:** 8 hosts distributed evenly across departmental subnets.

Protocols & Standards

- **IEEE 802.1Q (VLAN Tagging & Trunking):** Enables multiple VLAN traffic streams to travel across a single high-speed physical link between the switch and router.
- **Router-on-a-Stick:** A routing architecture utilizing sub-interfaces on a single physical router port to route traffic between disparate VLANs.
- **DHCP (Dynamic Host Configuration Protocol):** Centralized service on the router responsible for dynamically leasing IP configurations and default gateways to endpoints.
- **Port Security:** Layer 2 security mechanism restricting physical interface access to authorized MAC addresses only (configured with a maximum limit, sticky learning, and restrict violation mode).
- **Extended ACL (Access Control List - #100):** Advanced packet-filtering mechanism evaluating source/destination IPs, protocols, and control flags to enforce granular security zoning.
- **ICMP (Internet Control Message Protocol):** Diagnostic protocol utilized for network connectivity validation via `ping` commands and `echo-reply` tracking.

2. Network Topology & Addressing Scheme

- **Active Hardware Components:**
  - 1x Cisco 2911 Router (`R1`)
  - 1x Cisco 2960 Switch (`SW1`)
  - 8x End-Devices (evenly distributed across departments)
- **IP Subnet Architecture:**
  - **VLAN 10 (`IT_Dept`):** `192.168.10.0/24` | Gateway: `192.168.10.1` | Switch Ports: `Fa0/1` - `Fa0/4`
  - **VLAN 20 (`HR_Dept`):** `192.168.20.0/24` | Gateway: `192.168.20.1` | Switch Ports: `Fa0/5` - `Fa0/8`
  - **Trunk Link:** High-speed Gigabit connection (`SW1 Gig0/1` to `R1 Gig0/0` using IEEE 802.1Q encapsulation).

3. Core Configurations

Switch (`SW1`) Configuration

Plaintext

```
vlan 10
 name IT_Dept
vlan 20
 name HR_Dept

interface range fastEthernet 0/1 - 4
 switchport mode access
 switchport access vlan 10

interface range fastEthernet 0/5 - 8
 switchport mode access
 switchport access vlan 20

interface gigabitEthernet 0/1
 switchport mode trunk

interface range fastEthernet 0/1 - 8
 switchport port-security
 switchport port-security maximum 1
 switchport port-security violation restrict
 switchport port-security mac-address sticky
```

Router (R1) Configuration

Plaintext
```
interface gigabitEthernet 0/0
 no shutdown

interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0

interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0

ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1

ip dhcp pool IT_Pool
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8

ip dhcp pool HR_Pool
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
```

4. Engineering Troubleshooting Case Study: Resolving Stateless ACL Asymmetric ICMP Blockade

The Problem Statement & Symptom

- **Security Policy Objective:** Allow IT (`VLAN 10`) to initiate communication with HR (`VLAN 20`), while strictly denying HR from initiating any traffic toward IT.
- **The Incident:** Following the initial Extended ACL implementation, performing an ICMP echo request (`ping`) from an IT host (`192.168.10.2`) to an HR host (`192.168.20.2`) consistently resulted in **`Request timed out`**, despite an explicit `permit` rule for IP traffic from IT to HR.

Root Cause Analysis

1. **Stateless Nature of Cisco ACLs:** Standard and Extended ACLs in Cisco IOS operate without session state awareness. They filter packets independently based purely on packet headers (source, destination, protocol) without tracking connection states.
2. **The Failure Mechanism:**
    - When the IT host sent an ICMP echo request to the HR host, the router successfully matched it against the `permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255` rule, forwarding the packet into the HR subnet.
    - However, when the HR host processed the request and attempted to transmit the **ICMP echo-reply** back to the IT network, the router intercepted this return stream entering via the HR sub-interface (`Gig0/0.20`).

- Because the ACL evaluated the return packet independently, its source was the HR network (`192.168.20.0/24`) heading toward the IT network, which directly triggered the global explicit `deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255` statement. Consequently, the router dropped the return packet, causing the timeout.

The Engineering Solution

To bypass the stateless limitation for diagnostic protocols without compromising the core security isolation, an explicit granular exception rule was introduced to permit returning ICMP echo-reply packets originating from the HR subnet back to the IT subnet:

```
! Remove faulty ACL
no access-list 100

! Implement corrected Extended ACL with ICMP Echo-Reply Exception
access-list 100 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 100 permit icmp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
echo-reply
access-list 100 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 100 permit ip any any

! Re-apply ACL to HR sub-interface ingress path
interface gigabitEthernet 0/0.20
 ip access-group 100 in
```

5. Verification & Final Results

1. **DHCP Provisioning:** Verified via `ipconfig` on end devices, confirming proper dynamic allocation of subnet addresses and gateways.
2. **IT to HR Connectivity (Permitted):**
   - *Command:* `ping 192.168.20.2` (executed from IT host).
   - *Result:* `Reply from 192.168.20.2: bytes=32 time<1ms TTL=127` (Successful payload delivery and return tracking).
3. **HR to IT Connectivity (Denied):**
   - *Command:* `ping 192.168.10.2` (executed from HR host).
   - *Result:* `Reply from 192.168.20.1: Destination host unreachable` (Successfully blocked and dropped at the gateway by the ACL).